Project Overview

The project is structured as a **CMake-based multi-target solution** with separate engine and game layers.

- **EngineCore/** → core systems (ECS, Rendering, Physics, Audio, Input)
- **Game/** → gameplay logic, prefabs, levels, editor integration
- **Extern/** → third-party dependencies (GLFW, GLAD, ImGui, FMOD, stb, FreeType)
- **bin/** → build outputs (e.g., bin/Debug/Game.exe)
- **InstallerSample-V3/** -> Installer

The build system supports **Debug**, **Release and Installer** configurations and automatically copies required assets and DLLs after build.

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2. Build Configurations

Both configurations are set in the top-level CMakeLists.txt

Configuration	Optimization	Use Case
Debug	No optimization, full symbols	For gameplay debugging and editor use
Release	Optimized, stripped symbols	For final gameplay and performance testing
Installer	Optimized, No Editor	For Installer

- Compiler: **MSVC (Visual Studio 2022)**
- Warning level: /W4 on MSVC.
- Output executables are placed under bin/<Config>/Game.exe.

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3. Enabling or Disabling the Level Editor

Toggle Key

- **Press F1** → Opens or closes the **Editor Overlay** (ImGui-based).
- **Press F2** → Toggles the in-game GUI system (mute/exit buttons).

Modes

Mode	Description
Offline Mode	Editor hidden; game runs in pure runtime mode.
Real-Time Mode	Editor enabled; use ImGui to edit levels directly during gameplay.

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4. Current Key Interactions in the game

- **Key F11** - Toggle meshoutline debug + physics debugger
- **Key E** - Use In Game Ability
- **Key L -** Toggle Light
- **Shift Left Click -** Allows Multi-Select and Multi-Delete in Room Editor
- **Key F -** Interact with Enemies/Puzzles
- **Right Click -** Fire Projectiles
- **Key R -** Mutation Reduction
- **Key P -** Enable/Disable Paintbrush mode in Room Editor
- **Key F1** - enable IMGUI (as stated above)
- **Key F2** - enable in-game editor (as stated above)
- **Key F6** - Toggle Windowed/Full Screen
- **Key F3** - Rendering Layer Toggle
- **Key F4** - Physics Layer Toggle
- **Key F7** - Gui Layering Toggle
- **Key F8** - Debug Draw Layering Toggle
- **Key [** - Toggle Background Layer

- **Key]** - Toggle World Layer

Cheat Code:

- **Key 0**: On / Off God Mod
- **Key 1**: Set ability to default
- **Key 2**: Set ability to heal
- **Key 3**: Set ability to burrow
- **Key 4**: Set ability to projectile
- **Key 5**: Set mutation charge up to 0
- **Key 6**: Set mutation charge up to 100
- **Key 7**: Set the player hit points to 100
- **Key 8**: Set the player hp to 0

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5. Cmake Setup Guide

4.1 Install CMake

Download from the official site:
🔗 https://cmake.org/download/

Recommended:

- **Windows x64 Installer** (cmake-x.y.z-windows-x86_64.msi)

- **Minimum version:** 3.15 (required by top-level CMakeLists.txt)
  CMakeLists

After installation, verify installation:

cmake --version

4.2 Configure and Build the Project

1st method (faster but less convenient)

Open the root folder directly in **Visual Studio 2022 → "Open Folder"**.
Visual Studio will automatically detect your CMakeLists.txt and let you select **Debug** or **Release** from the toolbar without needing .sln files. From there, simply set main as the startup item and run.

2nd method (slower but convenient)

- Run the clean .bat file (optional)
- Run the build .bat file
- Run debug/release .bat according to user needs

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5. Debugging and Common Issues

Issue	Fix
❌ *"Source error" or project fails to load*	Right-click on the root CMake project in Visual Studio → "Delete Cache and Reconfigure".
❌ *Missing textures or shaders*	Verify that the Assets and shaders folders exist in the build output; CMake auto-copies them on build
❌ *Random Generation of map results in inaccessible pathway / affects gameplay*	Either method works: 1) clean build and build using the provided .bat file (takes longer) 2) simply go to build folder and delete labyrinth.txt, run it again to regenerate

6. Installer

Simply go to InstallerSample-V3, Installer and run the exe. You will have to accept the agreement, and optionally, create a shortcut.